

ARM-450: Design, Analysis and Pre-Print Verification of a 450 mm Printed 6-DOF Manipulator with Quick-Change End Effectors

Status — pre-hardware. Every result below is analytical or simulated. Nothing in this document has been printed or measured. Sections marked [AWAITING HARDWARE] are written with the measurement defined and the space left for the number; they are the whole point of publishing the analysis first, because a prediction recorded before the test is evidence and one recorded after it is not.

Date of this revision: 2026-08-22 · Verification state: 27 checks, 8 stages, 0 failures

Abstract

A 6-DOF serial manipulator of 450 mm reach was designed from a clean sheet for fused-filament printing in PLA, targeting two tasks: tracing a sinusoid on a vertical board as a repeatable accuracy demonstration, and a probe-and-drogue docking manoeuvre representative of an on-orbit refuelling interface. The arm masses 1.365 kg against a 1.450 kg ceiling and carries a 300 g payload allowance at the tool point.

The contribution is not the arm. It is the verification method: an eight-stage gate in which every criterion is computed from the exported CAD geometry rather than from the parameters used to author it, and in which the whole load suite is re-run for five payload configurations — bare flange through gripper-plus-225 g. That gate has so far found thirteen defects that survived visual inspection, three of which would have made the docking interface non-functional in plastic. All 27 checks currently pass.

The design is explicitly not flight hardware. Of sixteen subsystems compared against a space-grade equivalent, three survive the transition unchanged, two survive in principle, and eleven must be replaced. The kinematics and the joint architecture survive; the material, the actuators and everything that touches vacuum do not.

1. Motivation and scope

The work sits inside a maglev-floated chaser/target capture project (MASCOT), where the arm is the manipulation element. Two demonstrations were required:

1. Sinusoidal trace. A pen-equivalent tool follows a sine on a vertical board. This is a precision demonstration: the traced curve is a permanent record of the arm's repeatability, visible to the eye, and it exercises five of six joints simultaneously.
2. Docking. A probe enters a passive port and latches. This is a capture tolerance demonstration, and the interesting engineering is that it must succeed despite the arm being far less accurate than the interface would naively require.

1.1 Requirements

#	requirement	source	status
R1	≤ 450 mm chain, base to tool face	project brief	met, exactly 450.0 mm
R2	≤ 1450 g arm mass	project brief (hard)	met, 1365 g, 85 g spare
R3	300 g payload at the tool point	derived from task	met, worst case 75 g fitted
R4	6 DOF, wrist capable of arbitrary tool orientation	task	met
R5	Printable on a consumer FDM machine	project constraint	met, plain PLA, brass nozzle, no supports on load paths
R6	Tool change without tools, in seconds	usability	met, 3-lug bayonet + one thumbscrew
R7	Docking without a seventh actuator	mass and complexity	met, latch driven by J6
R8	Sine trace visibly accurate	demonstration	met in simulation, 0.055 mm rms at the tool tip

R1 is a constraint on the arm, not the system. With a gripper fitted the working point reaches 492 mm. This was raised explicitly and accepted as a design decision on 2026-08-22; it is now recorded as `SYSTEM_REACH_MAX = 495 mm` and checked, so a later and longer tool cannot quietly exceed what was agreed.

2. Architecture

2.1 One joint interface everywhere

The single most consequential decision in the design is that every joint uses the same mechanical interface: a Ø42.00 H-fit pocket, a 6806 (30×42×7) or 6706 (30×37×4) thin-section bearing, and a Ø30 shaft. One bore diameter, one shaft diameter, one clamp, one horn adapter, across all six axes.

The consequences compound:

- One fit tolerance to characterise on the printer, not six. A single 2.3 h test coupon gates roughly 19 h of parts.
- One spare part number. A failed bearing anywhere is the same bearing.
- The horn adapter, shaft clamp and aluminium tube are common parts, printed or cut six times.

The cost is that the wrist carries a bearing larger than it needs. That was measured, not assumed: substituting the 4 mm-wide 6706 at J4/J5/J6 saved 24 g, and moment stiffness

scales with bearing spacing squared and is independent of bearing width, so the narrower bearing cost nothing in stiffness once the spacing was preserved.

2.2 Links

Each link is a clamshell pair — a tongue half and a groove half — bolted along a seam on the neutral axis, forming a closed box section 47.5×29.0 mm with a 2.4 mm wall.

Closing the section is not cosmetic. An open C-section of the same envelope has a torsion constant roughly two orders of magnitude lower, and the payload hangs off-axis. The seam bolts are what make the section closed; they are M3 $\times$ 20, and an earlier M3 $\times$ 12 stopped 2.5 mm short of engaging.

2.3 Wrist

J4 (roll), J5 (pitch) and J6 (roll) are integrated into three printed housings rather than assembled from generic modules. J5 is a fork whose two cheeks are spaced as widely as the envelope allows, because moment stiffness goes as spacing².

J5 travel is asymmetric: -91° to $+49^\circ$, not $\pm 110^\circ$ as first specified. An exact-mesh sweep found the J4 housing striking the J6 output body over the excluded range. The URDF carries the narrowed limit and the collision checker parses limits from the URDF rather than holding its own copy.

2.4 Actuation

Six serial-bus servos: ST3250 at the shoulder (J2), ST3215 elsewhere. Selection is by gravity torque from a 60 000-pose sweep of the reachable configuration space, not by a single worst-case pose.

Torque reaches the printed structure through a horn adapter that clamps the servo output spline and pins into the joint shaft. Without it the torque path terminates at the servo's own horn screws.

3. Kinematics

3.1 Chain

world	→	base_link	→	link1	→	link2	→	link3	→	link4	→	link5	→	link6	→	tcp	→	tool	→	tool_tip
				J1 Z		J2 Y		J3 Y		J4 Z		J5 Y		J6 Z		fixed		fixed		
z (mm):				50		40		119		119		62		30		30		+42 (gripper)		

Total 450 mm base to tool face. tcp is the J6 face; tool_tip is where a fitted tool actually works.

tcp is not the working point, and conflating the two is a live error class. The sine trajectory drives the flange; with a gripper fitted, a board placed where the flange traces would be struck by the flange while the fingers were already 42 mm through it.

3.2 Inverse kinematics

Bounded least-squares (scipy.optimize.least_squares) on position plus tool-axis direction.

Two terms are non-obvious and both were arrived at by failure:

J6 is locked. A pen is rotationally symmetric about the tool axis, so J6 is pure redundancy. Leaving it free makes the solve under-determined and the solver wanders 135° between adjacent waypoints. Locking it converts an under-determined 6-DOF problem into an exactly-determined 5-DOF one.

Null-space regularisation on J4 and J6 only. When $J5 \rightarrow 0$ the two roll axes become collinear and only their sum is determined. Penalising all six joints equally suppressed the wander but destroyed accuracy ($0.02 \rightarrow 8.8$ mm rms) because it also fought the joints doing the task. Damping only the redundant pair costs nothing.

Orientation weight depends on whether a tool is fitted. With a bare flange, position and orientation are nearly independent and $w_{\text{ori}} = 0.05$ is right. With a tool, a tilt moves the tip, and at that weight the solver trades 21° of tool tilt for position error.

3.3 Verification of the kinematics

The chain was re-derived independently in modified (Craig) DH and cross-checked against ikpy:

check	worst error over 200 poses
FK: Craig DH == URDF chain	1.96×10^{-13} mm
FK: Craig DH == ikpy	2.17×10^{-13} mm
FK: URDF chain == ikpy	1.14×10^{-13} mm
Jacobian: velocity propagation == finite difference	1.81×10^{-5}
IK round trip, both solvers	$\leq 1.6 \times 10^{-11}$ mm median

7 of 7 pass, at double-precision rounding. The Jacobian is assembled by velocity propagation rather than differencing, which keeps it exact at the J4/J6 wrist singularity.

Agreement between the DH table and the URDF proves only that the table copies the URDF faithfully — errors included. The third implementation is what closes that: three routes sharing no code cannot agree on the same mistake. The derivation is in ARM450_KINEMATICS.pdf; the check is check_ikpy.py.

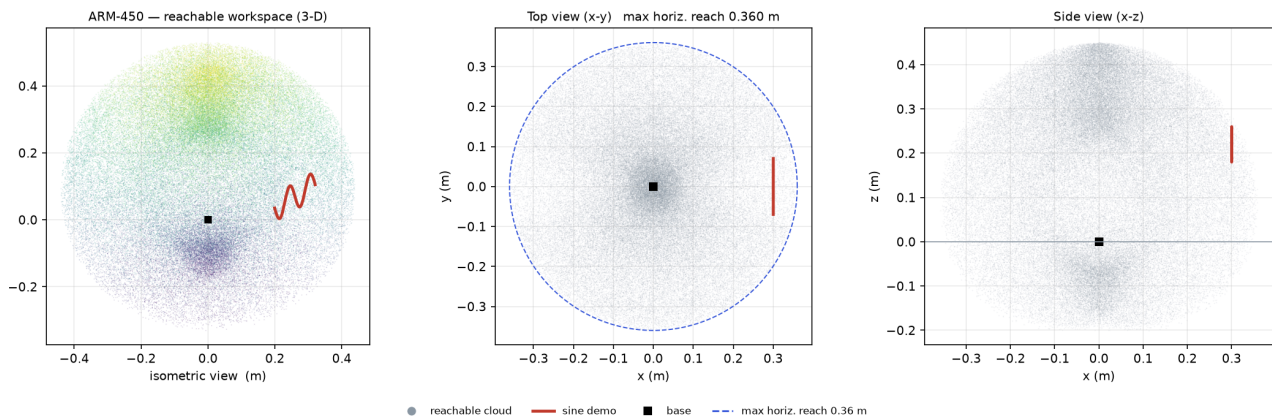
3.4 Workspace

Monte-Carlo forward kinematics, 200 000 poses sampled inside the URDF joint limits, binned into a 5 mm (r, z) grid and revolved.

fitted	reach past J6	max reach	horizontal	swept volume
bare face	—	450 mm	360 mm	188 L
pen	30 mm	480 mm	390 mm	239 L
gripper	42 mm	492 mm	402 mm	262 L
dock probe	36 mm	486 mm	396 mm	251 L

A gripper adds 39 % of swept volume for 42 mm of extension — volume grows roughly as the cube of reach.

ARM-450 — reachable workspace (verified vs URDF/FK) with sine demo overlaid



4. End effectors

4.1 Quick-change coupling

A three-lug bayonet plus one M3 thumbscrew. Push on, twist 30°, nip the screw. The adapter bolts to J6 once and stays.

Bolts were rejected because the J6 face has only three holes — the servo pocket eats the fourth quadrant — and starting three blind M3s under a wrist at every tool change is the wrong answer to a tool-change problem.

4.2 Gripper

Parallel jaws, rack and pinion, one SG90 (9 g). Parallel rather than scissor because the jaw faces stay square through the stroke; a scissor grips a cylinder at two points that migrate as it closes. A 90° V-groove in each finger locates round stock on four lines.

Stroke 31.4 mm per jaw over half a pinion revolution against a 30 mm design opening.

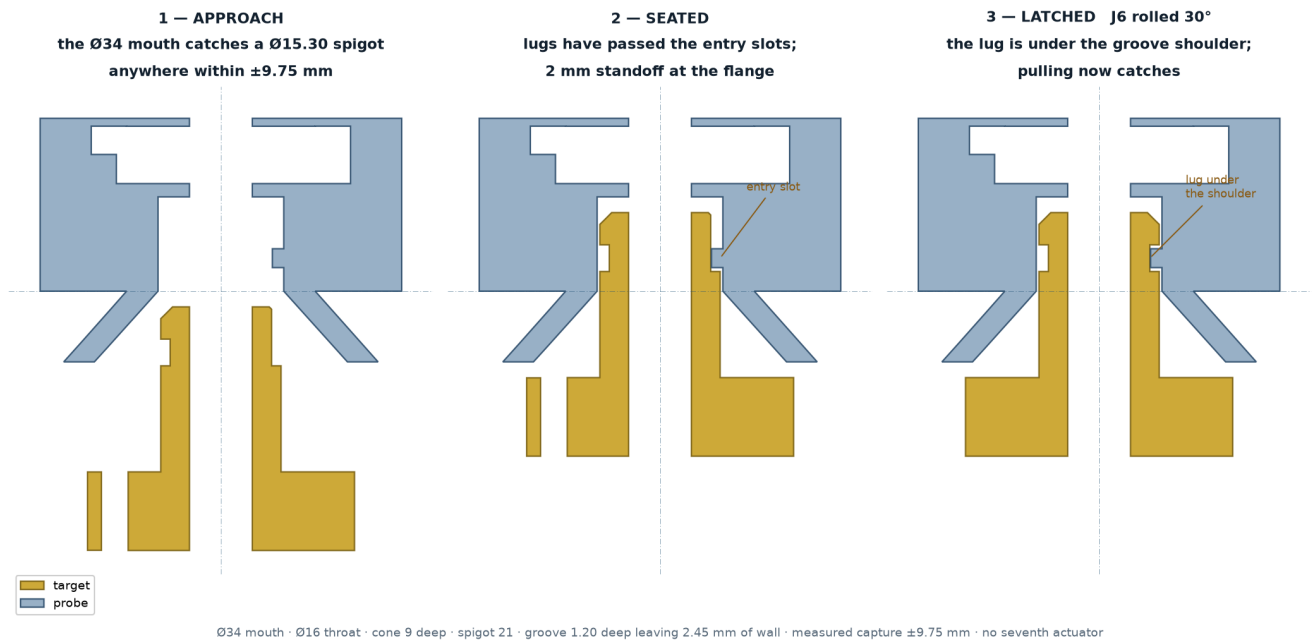
4.3 Docking probe

No seventh actuator. The latch is a bayonet and J6 turns it: insert, roll 30°, latched. J6 has $\pm 175^\circ$ and does nothing during a docking approach.

The design driver is capture tolerance, not accuracy. A $\varnothing 34$ mouth over a $\varnothing 16$ throat gives a measured ± 9.75 mm envelope — measured by sweeping the exported probe mesh sideways against the exported target mesh until they foul, not subtracted from two diameters. Expected arm precision is ≈ 2.2 mm, so the cone carries about 4× the error the arm can make, and about 2× the error it can make with worst-case backlash.

This is the design pattern worth taking from the project: docking was made feasible by geometry, not by making the arm accurate enough.

MASCOT DOCKING LATCH — section through a lug centre



5. Verification method

5.1 The principle

Every check reads the exported geometry — the STEP B-rep or the tessellated mesh — and not the parameters used to author it. The distinction is the entire value of the gate.

An early version checked that `BEARING_POCKET_D == 42.0` in the parameter file and passed 55/55 while three bearing pockets did not physically exist: a hollowing operation had removed the material the pocket was cut into, and subtracting a Ø42 pocket from an already- empty region succeeds silently and changes nothing.

A boolean that removes nothing does not fail. It returns the same solid, and every step downstream — render, drawing, mass, collision — treats the result as correct.

5.2 The eight stages

stage	question	criterion
1 Design check	Do the features physically exist?	STEP B-rep audit
2 FEA	Does anything yield?	p99 von Mises SF ≥ 3
3 Creep + fatigue	Does it fail slowly?	strain < 0.5 %/y, life > 1e8
4 Stiffness	Does the tool stay where it is put?	compliance ≤ 0.30 mm
5 URDF	Does the model match the machine?	parses, mass matches, meshes resolve
6 Path tracing	Does the actual trajectory work?	0 collisions, < 0.20 mm rms

stage	question	criterion
7 Simulation	Will the ROS side work on hardware?	trajectory published, ramp, rate
8 End effectors	Do the tools fit, does the latch latch?	29 interface checks + a simulated mate

5.3 Results, with and without an end effector

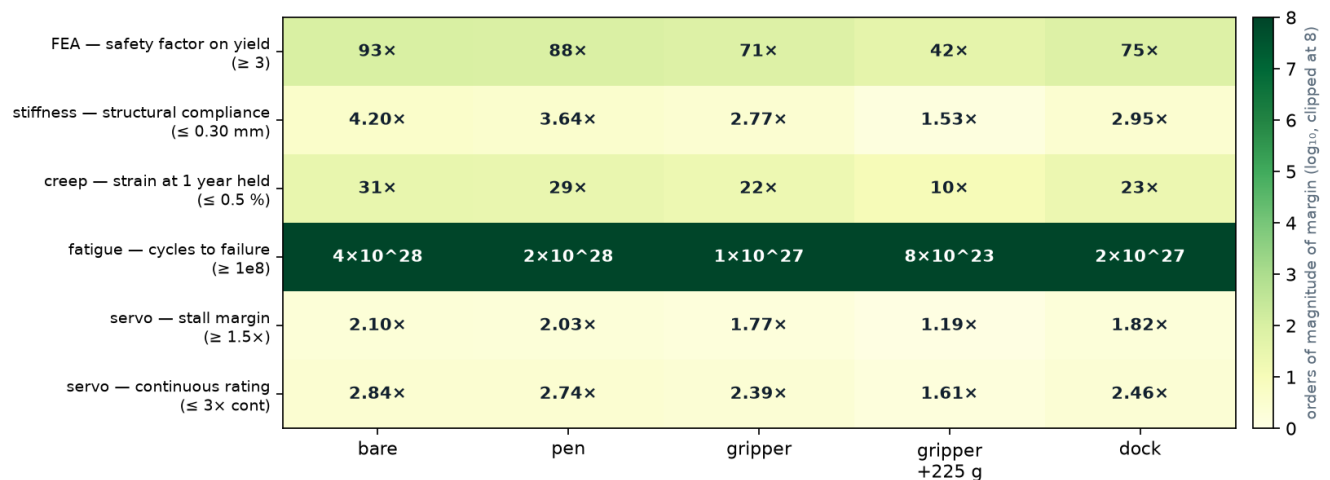
Every load analysis originally assumed a single number: 300 g at the tool point. That is a generous envelope but it says nothing about a fitted tool as such, because a tool is not only mass — it moves the load along the tool axis, and the lever arm is what sizes J2/J3 and sets the compliance at the point that matters. 75 g on a 42 mm stick is not the load case 75 g at the flange is.

The suite is therefore run for five configurations from one driver, so the "with gripper" answer cannot be a different vintage from the "bare" answer.

config	tool	held	J2 N·m	J3 N·m	σ MPa	SF	δ mm	creep %/y	cycles
bare	0 g	0 g	1.55	0.63	0.197	305	0.036	0.0159	3.9e36
pen	15 g	0 g	1.61	0.66	0.208	288	0.041	0.0172	1.8e36
gripper	75 g	0 g	1.85	0.82	0.257	234	0.054	0.0230	1.1e35
gripper + 225 g	75 g	225 g	2.74	1.39	0.437	126	0.197	0.0972	8.0e31
dock	62 g	0 g	1.79	0.78	0.245	245	0.051	0.0216	2.0e35

30 of 30 criteria pass — six criteria across five configurations.

ARM-450 — margin against every limit, with and without an end effector
the number is how many times better than the criterion, so 1.00x is exactly at the limit



Every cell passes. The two that are merely comfortable rather than enormous — servo stall margin and continuous rating in the gripper+225 g case — are the ones worth watching on hardware; the structural margins are so large because this arm is stiffness-driven and servo-limited, not strength-driven.

The structural margins are enormous (46x to 102x on yield) and this is not a claim of excellence. It is a statement that the arm is stiffness-driven and servo-limited, not

strength-driven. The two margins that are merely comfortable — servo stall at 1.19× and continuous rating at 1.61× in the worst configuration — are the ones to watch on hardware.

5.4 Precision budget

config	structural	0.5° backlash	combined
bare	0.036 mm	1.88 mm	1.88 mm
gripper	0.054 mm	2.24 mm	2.24 mm
gripper + 225 g	0.098 mm	2.24 mm	2.24 mm

Backlash dominates by roughly 20× in every configuration. No change to the printed parts moves the number. Improving accuracy means joint-side encoders, not a stiffer arm — and that conclusion is invariant to whether a tool is fitted.

6. Trajectory and simulation

IK is solved offline; the ROS 2 node replays the solved file. A least-squares solve per waypoint inside the node would block the executor.

Shipped trajectory: 240 waypoints, board at 300 mm (flange) / 342 mm (gripper tip), 40 mm amplitude, 2 cycles, 140 mm span, trace centre 220 mm.

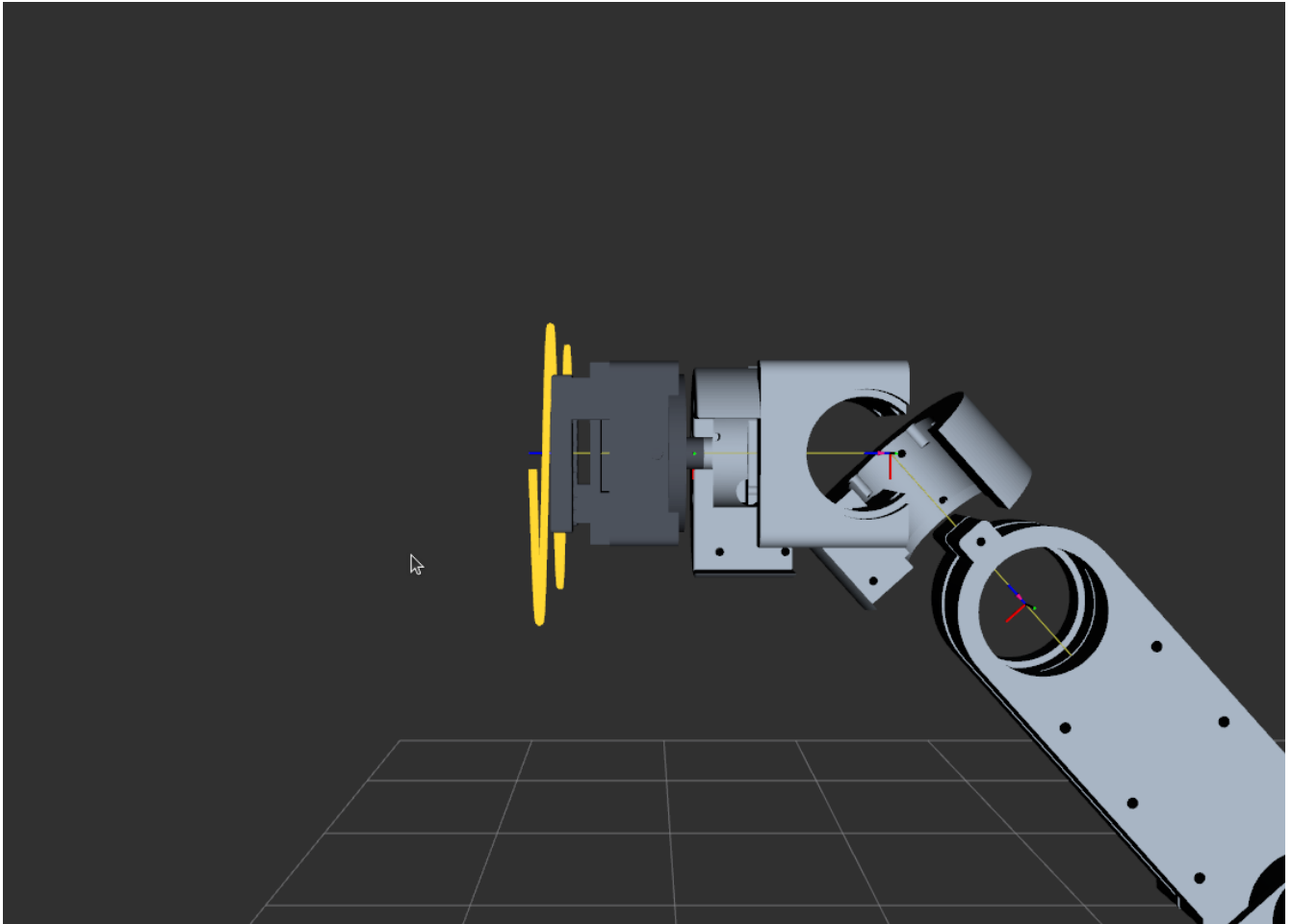
- position error 0.014 mm rms, 0.217 mm max (flange)
- tool-tip error 0.055 mm rms, 0.834 mm max
- tool tilt 0.093° rms
- largest joint step between waypoints 2.09°
- 0 collisions in 240 waypoints on exact mesh geometry; 0/60 with either tool fitted

Three hardware-readiness features exist because simulation hid the need for them:

Approach ramp. The node published waypoint 0 on its first tick — invisible in RViz, a full-speed slam on real servos. Now a cosine ramp from the measured start pose.

Ping-pong playback. Modular wrapping from the last waypoint to the first is a discontinuity, not a loop: J4 jumps 43.3° in one 40 ms tick and the tool teleports 139.8 mm.

A real trajectory topic. /joint_states is a visualisation stream a controller will not follow. The solved path is also published once as a latched 240-point JointTrajectory.



7. Comparison with a space-grade manipulator

The brief described the arm as "space grade with minor adjustment". That is true in one specific sense and false in every other, and the distinction is worth stating precisely.

What survives a move to flight hardware, and what does not

SUBSYSTEM	ARM-450 (this build)	SPACE-GRADE EQUIVALENT	
Kinematics & DOF layout	6R, 450 mm, offset-free wrist	identical	
Joint architecture	Ø42 pocket · 6806 · Ø30 shaft, one part number throughout	same scheme, flight bearings	
Fastening scheme	M3 steel, heat-set brass inserts	same layout; locked, staked, vented	
Structural material	plain PLA, E ≈ 3.5 GPa, Tg 60 °C	Al 6061-T6 or CFRP, E 70 GPa	
Mass	1.584 kg assembled (1.34 structure + 0.24 fasteners)	≈1.13 kg in Al 6061, ~10x stiffer	
Actuators	ST3215/3250 hobby serial servos, plastic gear train	BLDC + harmonic drive, vacuum-rated, rad-tolerant	
Feedback	servo-internal pot, no joint encoder	joint-side resolver / optical encoder	
Precision	≈2.2 mm, backlash-dominated	< 50 µm, encoder-closed	
Lubrication	wet lithium grease in 6806	dry film MoS ₂ / PFPE, vacuum-stable	
Thermal range	0–50 °C; Tg 60 °C is the wall	–120 to +120 °C cycling, CTE-matched	
Vacuum compatibility	none — PLA outgasses, trapped volumes unvented	TML < 1 %, CVCM < 0.1 %, vented	
Radiation	none	TID-rated parts, latch-up immune	
Qualification	analysis + bench only	vibration, shock, thermal-vac, EMC	
Redundancy	single string	dual-wound motors, redundant sensing	
Docking interface	3-lug bayonet, J6-driven, ±9.75 mm capture	same principle; softer capture, latch preload, fluid seal	
Cost	consumer printer + hobby servos	3–4 orders of magnitude higher	

 survives the transition unchanged
  survives in principle, needs rework
  does not survive — must be replaced

Of sixteen subsystems: 3 survive unchanged, 2 survive with rework, 11 must be replaced.

What survives

The kinematics. A 6R chain with this joint layout and these limits is unaffected by what it is made of. The URDF, the IK and the workspace all transfer directly.

The joint architecture. One bore, one shaft, one bearing family across all six axes is good practice at any grade; only the bearing's part number and lubricant change.

The fastening scheme. Bolt patterns and load paths transfer. Flight versions get locking compound, staking and vented screws — but in the same places.

What does not

The material is the wall. PLA+CF has a glass transition at ~60 °C; a sun-facing surface in LEO exceeds that. It outgasses, disqualifying it near optics or seals. Re-made in Al 6061-T6 the same geometry masses ≈1.13 kg — lighter than the printed version — and is about 10x stiffer, which is the honest argument for the swap: it is not a compromise.

The actuators are the second wall. Hobby serial servos have plastic gear trains, wet grease, no radiation tolerance and no joint-side feedback. This single choice sets the 2.2 mm precision figure.

Vacuum compatibility was never designed in. The printed parts contain closed internal volumes with no vent path — a trapped-volume problem that would need geometry changes, not material changes.

Nothing has been qualified. No vibration, shock, thermal-vacuum or EMC testing.

The honest summary

ARM-450 is a kinematic and architectural prototype of a space manipulator, and a functional demonstrator of a docking interface. It is not a flight article and no part of it is on a path to becoming one without replacement of the material, the actuators and the electronics. What it does transfer is the geometry, the joint scheme, the docking principle — and the verification method, which is grade-independent.

8. Defects found by the gate

Thirteen defects survived visual inspection and were caught by geometric interrogation. Selected, with the class of error each represents:

#	defect	class
1	Three bearing pockets absent; a hollowing op had removed the material they were cut into	boolean no-op
2	tool_dock exported as four solids — three latch lugs floating 5.5 mm clear of the bore	union without contact
3	dock_target groove cut 3.30 mm into a 1.65 mm wall, severing its own spigot tip	dimension not checked against what it cuts
4	Capture cone deeper than the spigot was long — probe and target could not mate at any depth or angle	relationship between two parts, in neither drawing
5	Rack pitch line 1.20 mm outside the pinion PCD — the gear mesh did not touch	looks right in a render
6	J6 pilot bore asserted on a drawing but absent from the part	drawing ahead of geometry
7	Steel shaft quoted at 93 g, actually 222 g; six would exceed the entire mass budget	per-unit figure never multiplied out
8	Seam screws 2.5 mm too short to reach the seam	grip length $\neq$ boss height
9	Sine path grazing the base pedestal on 30 % of waypoints while reporting 0.19 mm accuracy	accuracy is not clearance

#	defect	class
10	Playback wrap discontinuity: 43.3° in one tick	a loop that is not a loop
11	RViz loading a primitive URDF with unresolvable mesh paths	silent fallback
12	The URDF had no tool link at all — every RViz run showed a bare flange	verified in one model, displayed from another
13	Sine traced by the flange, not the tool tip — a 42 mm error on hardware	controlled point ≠ working point

Three documentation defects of the same family were also found: a drawing banner asserting a defect that had been fixed, every drawing quoting mass at a different infill than the budget used, and a PDF footer describing the wrong document on seventeen of eighteen files.

The pattern across all of them: nothing failed loudly. Each produced a plausible artefact — a render, a number, a passing check — that was wrong.

9. [AWAITING HARDWARE] Planned measurements

The analysis above is complete and self-consistent. It is not evidence about the physical arm. This section states what will be measured, what is predicted, and what would falsify the prediction — written before the measurement.

9.1 Fit coupon — gates everything

item	value
Print	fit_coupon.stl, 70 g, ≈2.3 h — both bearing sizes
Measures	Ø41.95 / Ø42.00 / Ø42.05 pockets, three insert-boss sizes, a seam sample
Prediction	Ø42.00 gives a firm thumb press; the printer's real bore ≈0.1 mm undersize
Falsified if	the bearing drops in free, or will not start
Consequence	BRG_FIT changes and every pocketed part is reprinted — ≈19 of 21 print hours

Result: _____

9.2 Servo backlash — the dominant error term

item	value
Method	Dial indicator at the tool tip; drive each joint to a pose, reverse, measure lost motion
Prediction	0.3–1.0° per joint; combined tip error 1.9–2.2 mm
Falsified if	measured tip error < 0.5 mm (structure-limited after all) or > 5 mm
Consequence	Sets whether the sine demo is visibly accurate and whether docking needs the full cone

Result: _____

9.3 Sine trace accuracy

item	value
Method	Trace on paper at 342 mm; scan; fit against the commanded curve
Prediction	1–3 mm rms, backlash-dominated, against 0.055 mm simulated
Falsified if	rms < 0.5 mm — the backlash model is wrong

Result: _____

9.4 Docking latch, bench

item	value
Method	Bolt dock_target to the bench; insert tool_dock by hand; roll J6 30°
Prediction	Latches and releases repeatably; withdrawal blocked when latched
Falsified if	the lugs bind, or the latch releases under axial load
Note	Needs no arm, no fuel and no flight software — the cheapest possible proof

Result: _____

9.5 Static deflection

item	value
Method	Arm horizontal at full extension; hang 300 g; dial indicator at the tip
Prediction	0.098 mm structural

item	value
Falsified if	> 0.30 mm — the closed-section assumption is wrong, i.e. the seam is not closing

Result: _____

9.6 Joint torque and thermal

item	value
Method	Servo bus feedback for current; IR thermometer on the J2 case after 3 min of tracing
Prediction	J2 peak 2.74 N·m against 4.90 stall; case 44 °C after 3 min, 105 °C only at steady state
Falsified if	J2 stalls, or the case exceeds 60 °C (PLA Tg) during a trace

Result: _____

9.7 Mass

item	value
Method	Scale, assembled, no tool
Prediction	1365 g ± 40 g
Falsified if	> 1450 g

Result: _____

10. Limitations

Stated plainly, because a verification document that does not is not one.

1. No hardware. Every number is analytical or simulated.
2. FEA is part-wise, not assembly-level. Joint stiffness, bolt preload and contact are not modelled. Part margins are 46–102×, so this is unlikely to be limiting, but it is a real gap.
3. Backlash is assumed, not measured, and it dominates the error budget — so the headline precision figure is the least evidenced number in the document.
4. Servo case-screw positions are from a datasheet, not from a measured servo.
5. Creep uses a Findley fit two orders of magnitude below its calibration range. The absolute strain is tiny but the extrapolation is long.
6. Print anisotropy is not modelled. Layer adhesion is typically 40–70 % of in-plane strength; orientation notes exist per part but the FEA is isotropic.
7. The docking mate is verified as rigid-body geometry, without friction, compliance or misalignment dynamics.
8. Thermal is steady-state or lumped, with no transient FE model.

11. Conclusions

1. A 450 mm 6-DOF printed manipulator meeting a 1450 g ceiling with a 300 g payload allowance is achievable in carbon-filled PLA, at analytical margins of 46–102× on yield and 0.098 mm structural compliance in the worst configuration.
2. The arm is servo-limited, not structure-limited. Backlash exceeds structural deflection by $\approx 20\times$, in every payload configuration. Accuracy work belongs in feedback, not in the printed parts.
3. Docking is feasible despite that, because capture tolerance was designed as geometry: a ± 9.75 mm cone against ≈ 2.2 mm of arm error. Making the interface forgiving is cheaper than making the arm accurate.
4. A single joint interface across all six axes compounds into fewer tolerances, one spare, and a one-coupon qualification path.
5. Verification must interrogate the exported geometry. Thirteen defects survived visual inspection and passing parameter checks; three would have made the docking interface non-functional. The generalisable rules are collected in ARM450_LESSONS.pdf.
6. The design is a kinematic and architectural prototype of a space manipulator, not a flight article. Three of sixteen subsystems survive the transition unchanged.

12. Reproducing this work

```
git clone <repo> && cd arm450_design
python3 cad/end_effector.py      # rebuild the tool parts
python3 cad/volumes.py          # exact STEP volumes
python3 verify_configs.py       # the load suite, all 5 configurations
python3 preflight.py            # the full 8-stage gate
python3 make_commands.py        # regenerate the command reference
```

Supporting documents: ARM450_REPORT.pdf (full engineering report), ARM450_LESSONS.pdf (errors and the rules they produced), ARM450_DRAWINGS_ISO.pdf and ARM450_DRAWINGS_3VIEW.pdf (18 dimensioned parts), ARM450_BUY.pdf (procurement and print order), ARM450_COMMANDS.pdf (every command), ARM450_PREFLIGHT.pdf (the gate), ARM450_END_EFFECTORS.pdf, ARM450_WORKSPACE.pdf, ARM450_URDF.pdf, ARM450_KINEMATICS.pdf.

Appendix A — Mass budget

group	mass
Printed structure (PLA+CF, 90 % effective fill)	585 g
Servos, 6 ×	360 g
Bearings, 12 ×	134 g
Aluminium shafts, 6 ×	84 g
Horn adapters, fasteners, inserts	202 g
Arm total	1365 g

group	mass
Ceiling (hard)	1450 g
Spare	85 g
Gripper + adapter + SG90 (payload, not arm)	75 g

Appendix B — Verification criteria

criterion	limit	rationale
FEA safety factor	≥ 3	standard for a non-critical printed structure
Structural compliance	≤ 0.30 mm	below the servo backlash floor, so structure never dominates
Creep strain, 1 year held	≤ 0.5 %	below the elastic strain at working stress
Fatigue life	$\geq 1e8$ cycles	no polymer endurance limit, so a life target is used
Servo stall margin	$\geq 1.5\times$	covers dynamic overshoot on a static analysis
Servo continuous rating	$\leq 3\times$	tracing is transient; the thermal time constant is 9 min
System reach with tool	≤ 495 mm	the accepted envelope
Mass	≤ 1450 g	hard project ceiling